UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en Inligtingtegnologie / Lefapha la Boetšenere, Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 1: Needs Analysis

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh

Quick Recap/Questions arising from Previous Lecture

An overview of the module was the focus of unit 0. This covered discussions:

- revolving around the primary goal of the module i.e. entrepreneurial mindedness through business systems design and management
- on assessment opportunities in the module
- about pass conditions (covering marks and competence)
- on different knowledge areas of the module
- about group formation for the semester project and the list goes on.

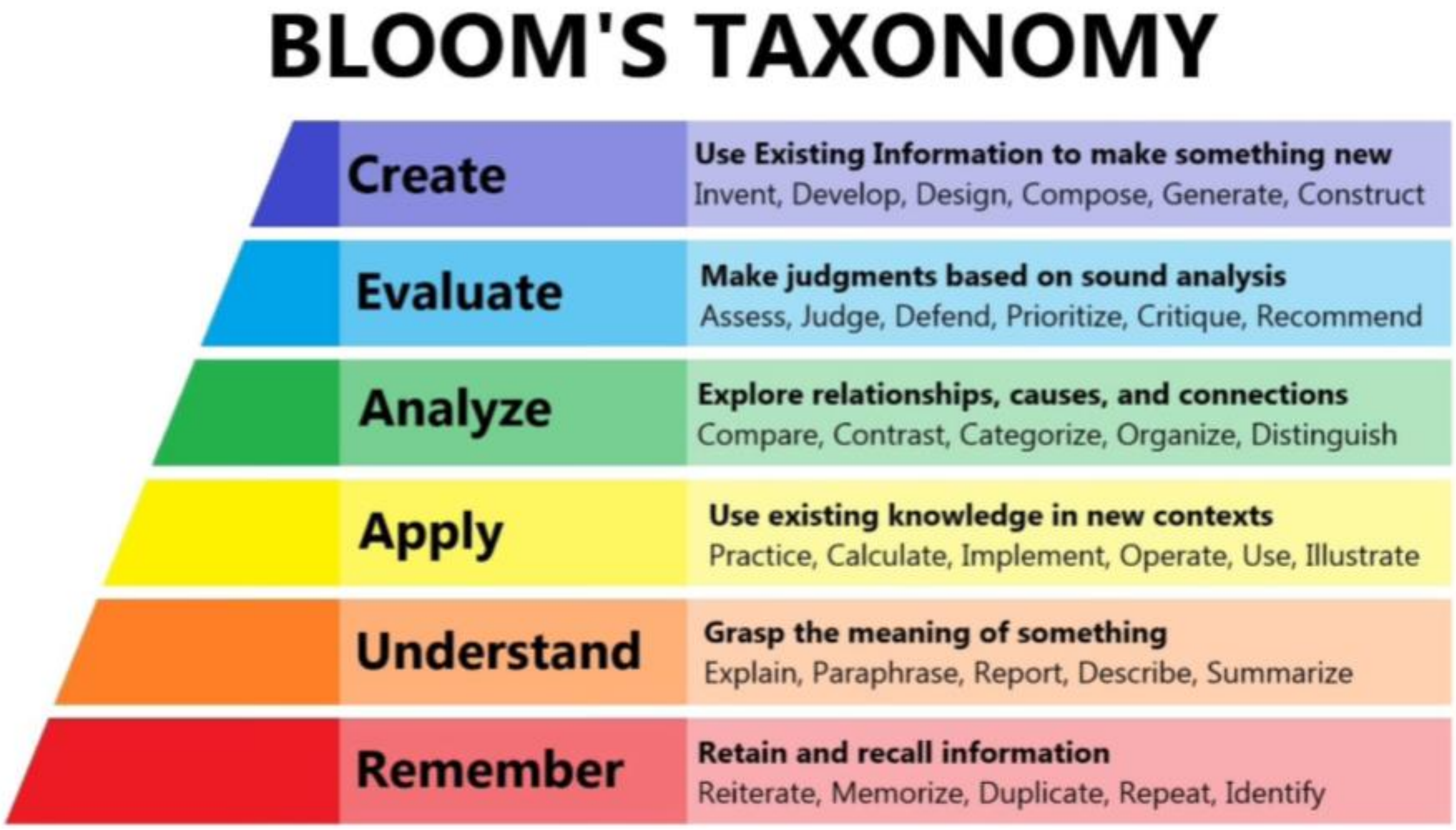
Expected Learning Outcomes for the Current Lecture

The expected learning outcomes herein include:

- Having a clear **understanding,** and being able to **apply, analyse**, and **evaluate** the concept of "Needs Analysis" and its subsidiary elements in respect of business systems design and management.

- Being able to **apply** this knowledge to the conceptual design of a new business system in the real world.

BLOOM'S TAXONOMY & EXPECTED LEVELS of COGNITION



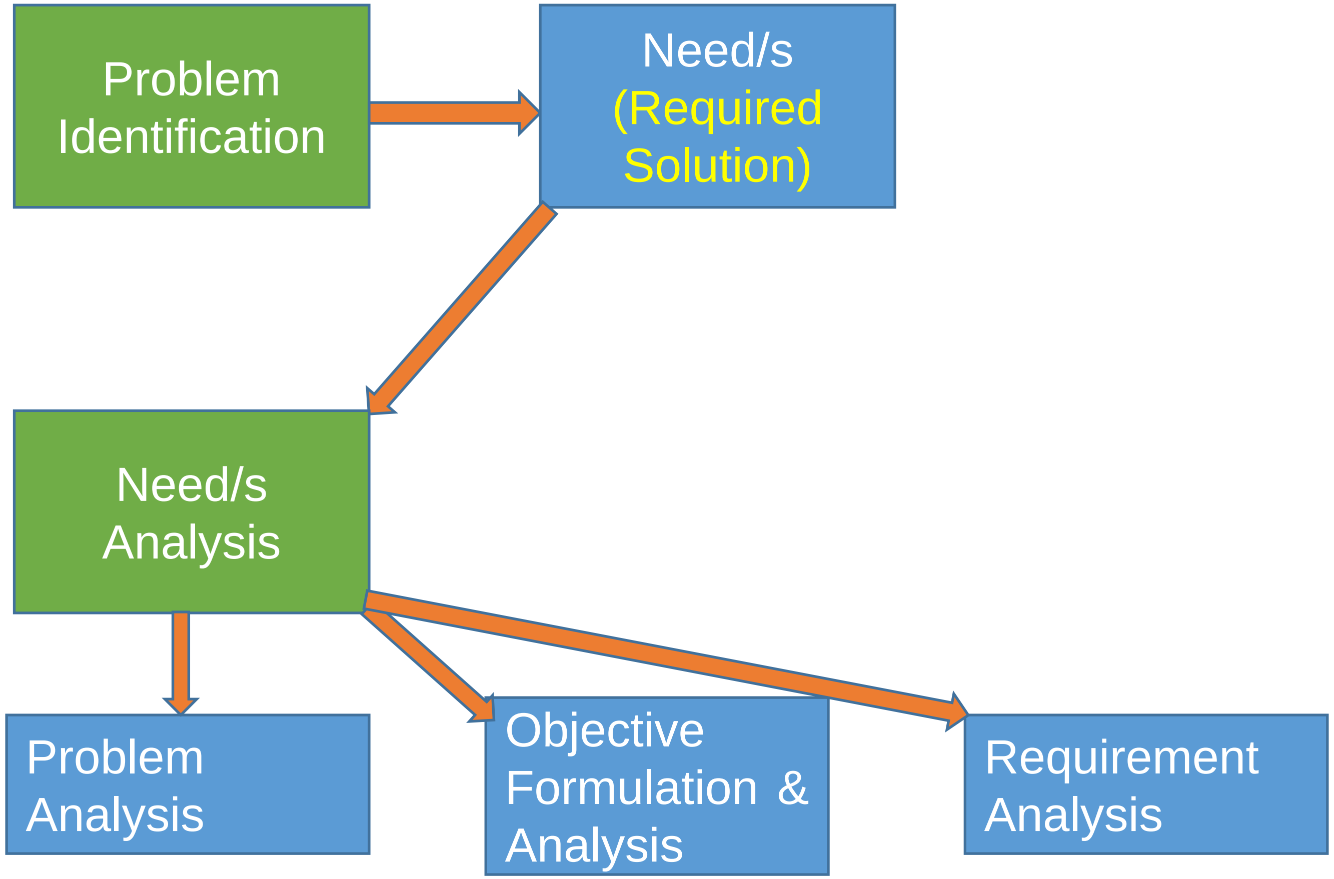
Presentation Outline

- Explain the terms “Needs” and “Needs Analysis” in Systems Design
- Problem Identification for Systems Design
- Definition of Objectives
- Requirements Analysis and Drivers of Systems
- Creation of Functions from Objectives
- Identification of Physical Embodiments for Functions
- Validation of a System’s Design

Knowledge Area

- Business Systems Design & Management (BSDM) LINKED TO THE SYSTEMS ENGINEERING BODY OF KNOWLEDGE

NEEDS explored Schematically



Problem Identification

- Gaps identification. Problems are always negatively viewed and presented. Problems are hardly ever positive in nature.

Definition of Needs

- A need in the context of systems design is anything that is REQUIRED for the functionability, operationability and/or performance enhancement of a system.
- The terms “REQUIRED” and “DESIRED” are respectively affiliated to “needs” and “wants”.
- Needs analysis is concerned with the compartmentalisation of an identified need into its constituent member elements including: problem identification, objective identification and analysis, functional elements, system drivers , physical elements and design validation.

Problem Analysis

- Analysis of the identified gap/s for needs specification

Objectives Formulation and Analysis

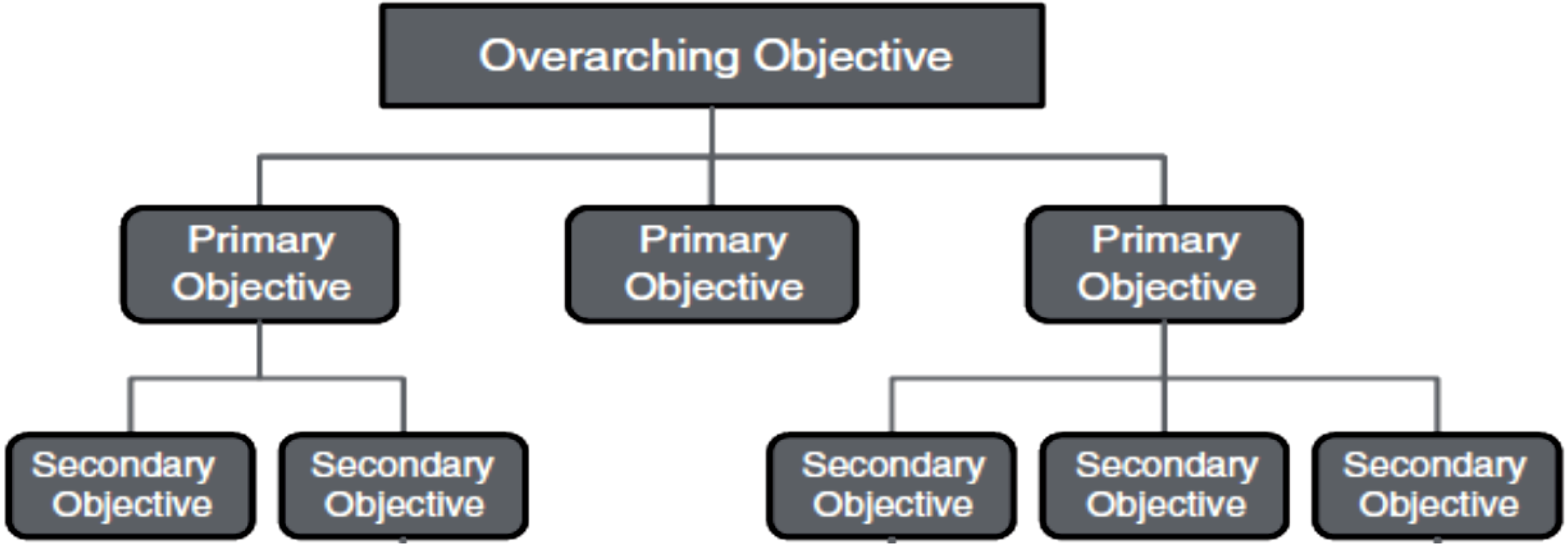
- **a** process of **developing and refining** the objectives of a system.
- the product of this effort is an objective tree
- where a single or small set of top-level objectives are decomposed into a set of primary and secondary objectives.

- **In determining whether an objective needs further decomposition, one should ask a couple of questions:**

Objectives Formulation and Analysis Contd….

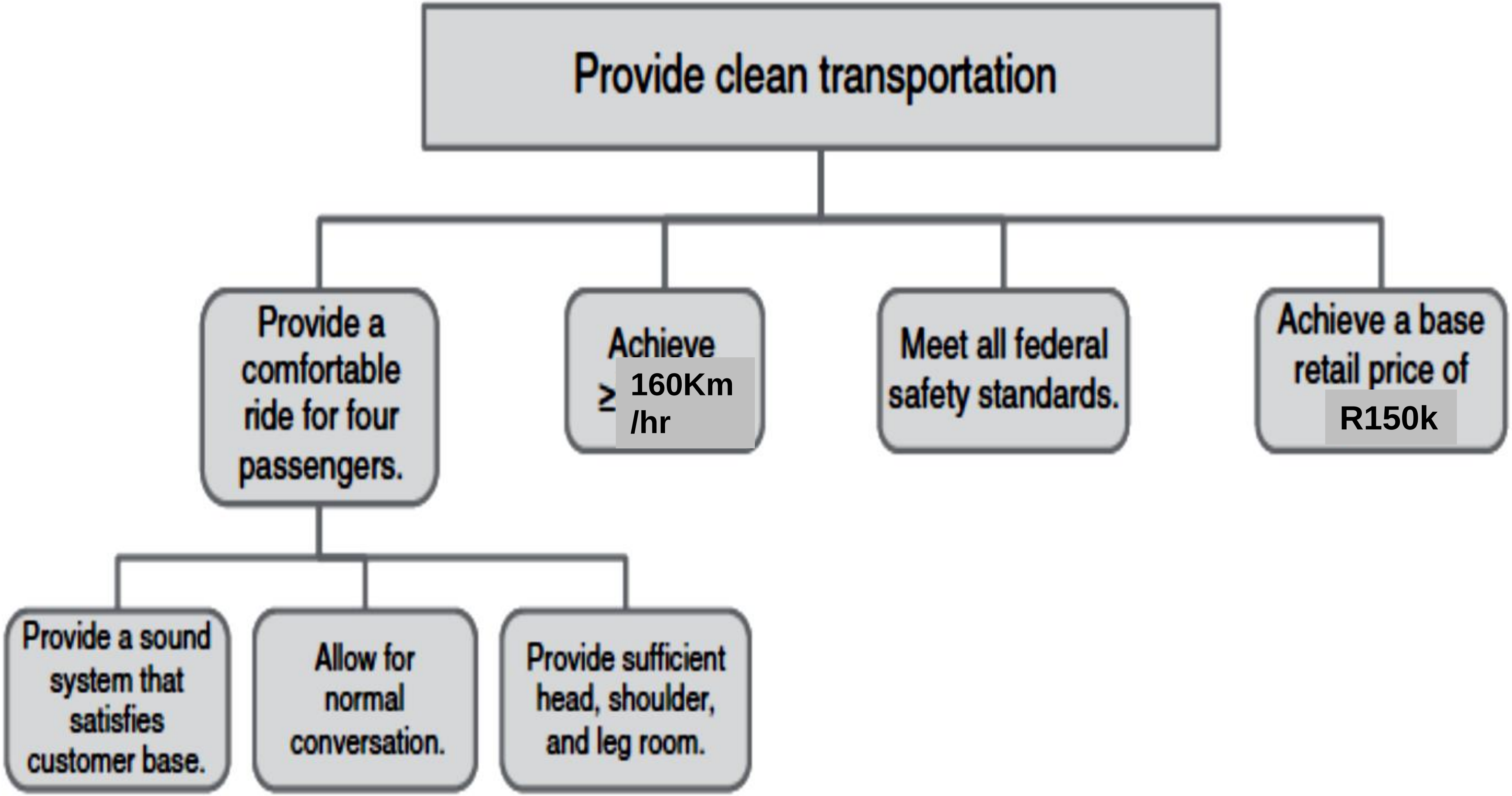
- Does the objective stand on its own in terms of clarity of understanding?
- Is the objective verifiable?
- Would decomposition lead to better understanding?
- Are requirements and functions readily implied by the objective?

Objective Tree Diagram



General Structure of an Objective Tree

Objective Tree Diagram



An Example of a Passenger Vehicle Objective Tree

Requirements Identification & Analysis

Requirements are system design considerations/inputs. These can include:

- Material types to be selected
- Skill set of human resources to operate machine etc
- Interfacing elements
- Surface roughness of material
- Choice of communication signal

Drivers for the Choice of System Requirements

These can include:

- assessment of deficiencies of the current system [operational deficiency]
- obsolescence
- technically feasible approach
- affordable cost
- acceptable level of risk
- potential market opportunity
- assessment of competitive products

Functional Analysis

- **Typical activities include**
  - translating operational objectives into functions that must be performed
  - allocating functions to subsystems by defining functional interactions and
  - organizing them into a modular configuration.

The general product of this activity is a list of initial functional requirements

Functional Block Diagram…Contd…

- **Example: Eleven functions were identified as listed below:**
- **Input Functions**
  - Accept user command (on/off)
  - Receive coffee materials
  - Distribute electricity
  - Distribute weight

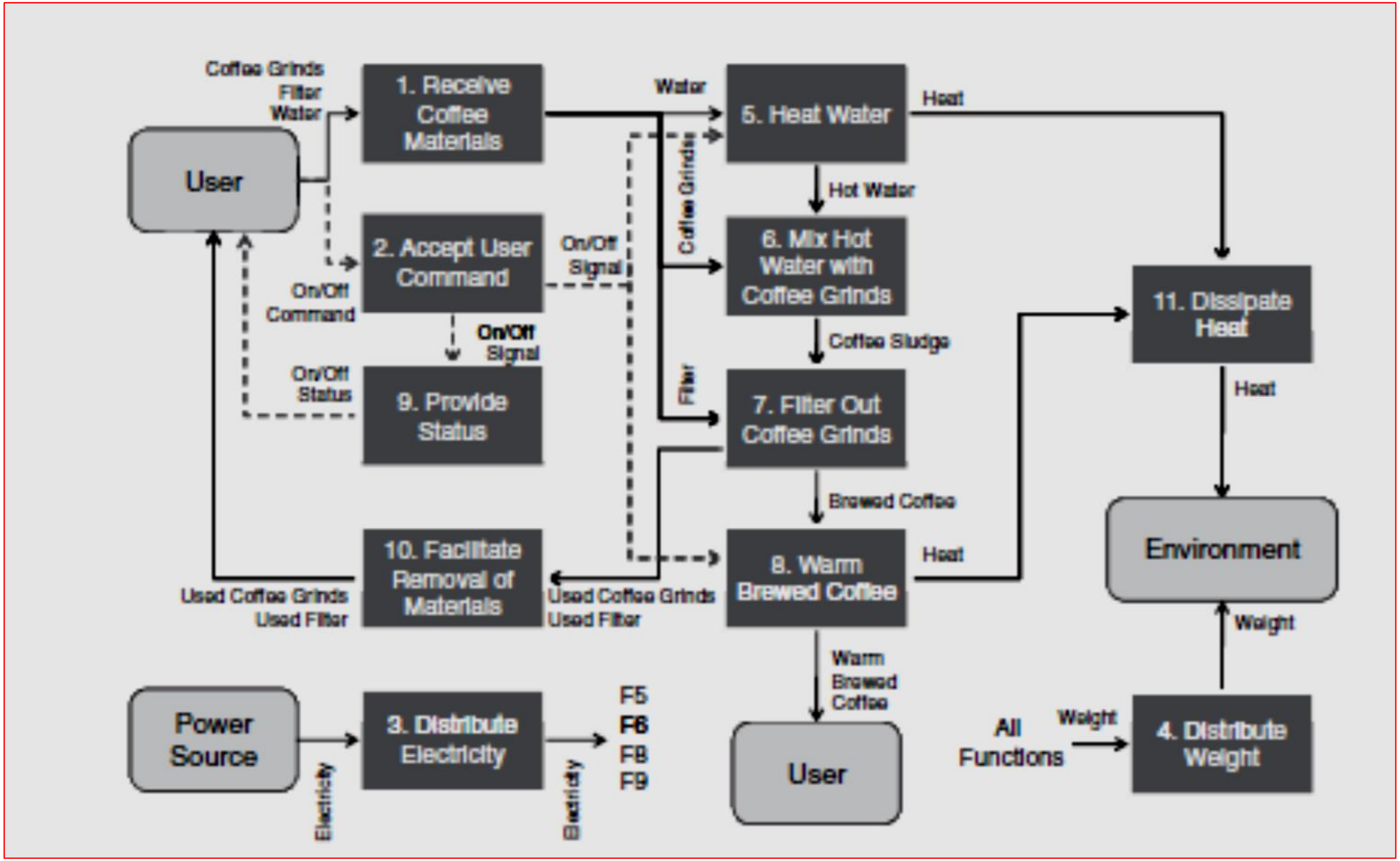
Functional Block Diagram…Contd…

- **Transformative Functions**
  - Heat water
  - Mix hot water with coffee grinds
  - Filter out coffee grinds
  - Warm brewed coffee

Functional Block Diagram…Contd…

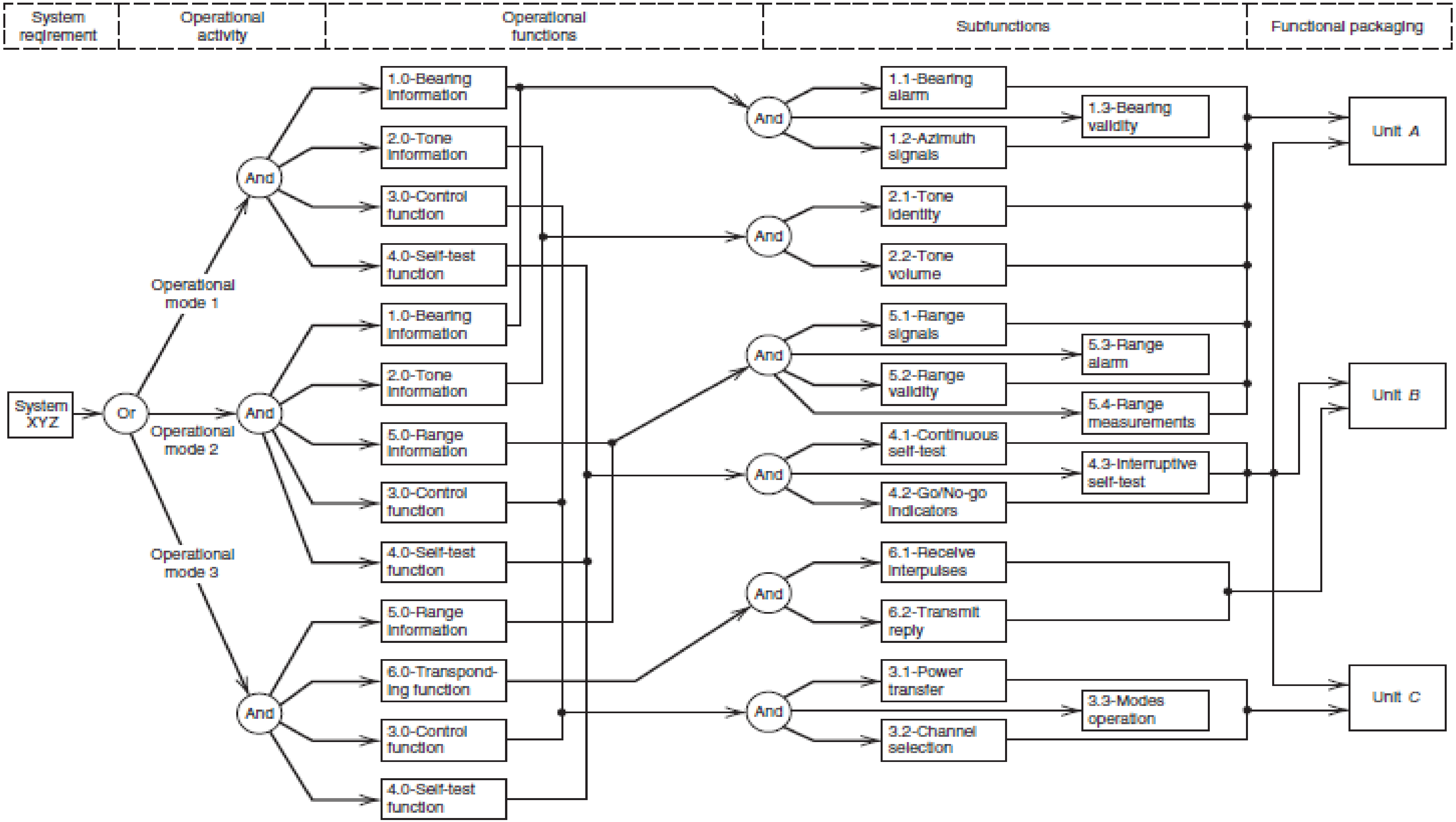
- **Output Functions**
  - Provide status
  - Facilitate removal of materials
  - Dissipate heat
- **Three external entities were also identified viz:**
  - the user,
  - a power source (assumed to be an electrical outlet), and
  - the environment

Functional Block Diagram for a Standard Coffee machine



Part two [Systems Engineering Principles and Practice, Second Edition; Author: Alexander Kossiakoff et al.] Figure 8.4

Functional Analysis Leading to System Packaging



System Engineering Management, Fifth Edition, Benjamin S. Blanchard & John E. Blyler, Chapter 2: Figure 2.22

Feasibility Definition [Physical Definition]

- Typical activities include:
  - visualisation of subsystem implementation
  - visualising the physical nature of subsystems
  - defining a feasible concept in terms of capability and estimated cost by varying (trade-offs).

The general product of this activity is a list of initial physical requirements

Feasibility Definition [Physical Definition]

Definition of Feasible Concepts

- **This covers but not limited to the following:**
  - discussion of the development process/strategy
  - anticipated risks
  - design approach
  - evaluation methods
  - production issues
  - concept of operations
  - some cost information on system development and production

Needs Validation Contd…

Validation is a process of **authentication** through a thorough investigation and analysis of a design process, product or service amongst others.

E.g.: There are two people in the hall. To validate this information, some actions must be carried out.

i. Go to the hall and
ii. Count the number of persons in the hall

Needs Validation Contd…[Operational Effectiveness]

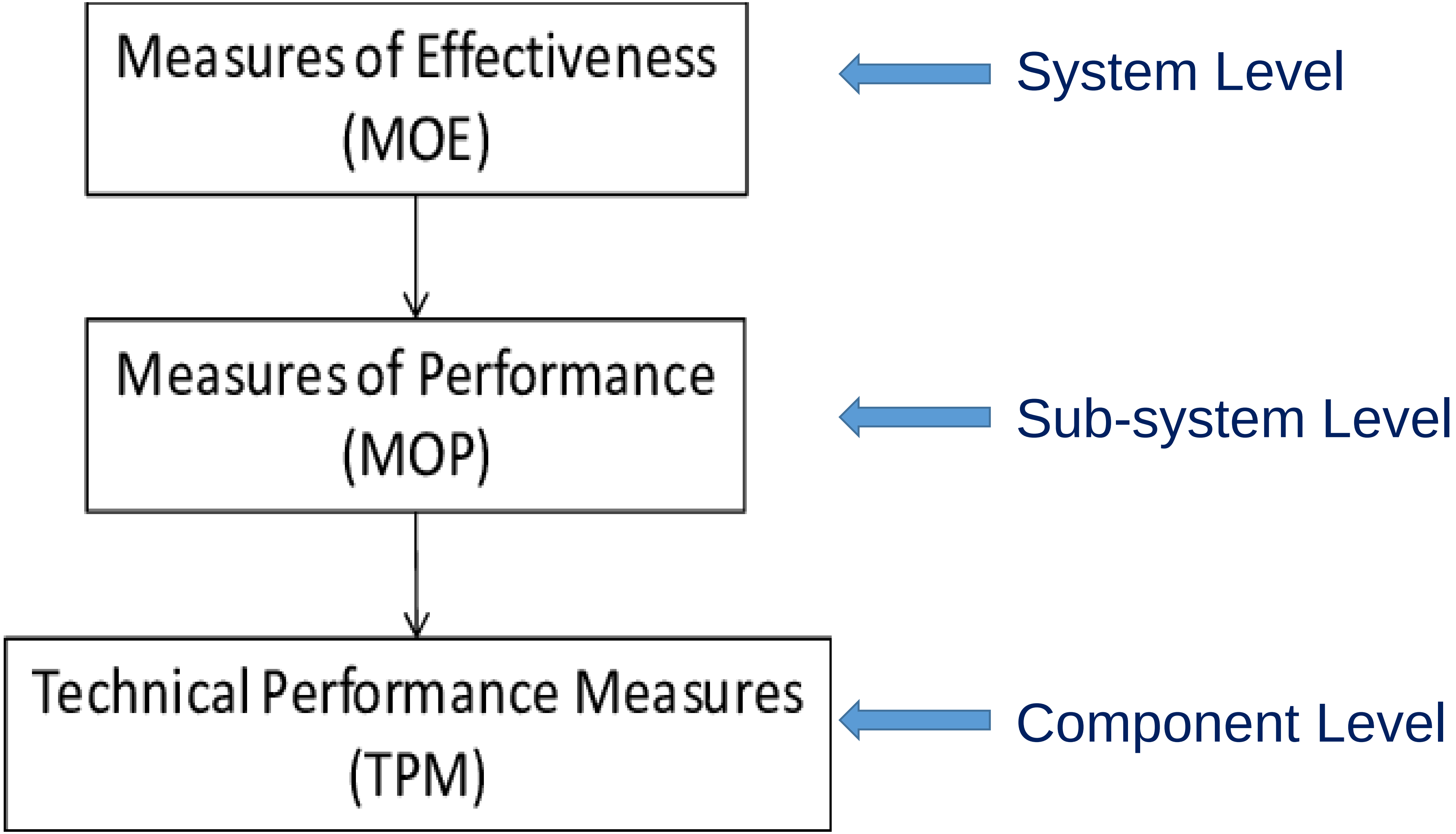
- **Measure of Effectiveness (MOE)**
  - a qualitative or quantitative metric of a system's overall performance.
  - an MOE always refers to the system as a whole.

- **Measure of Performance (MOP)**
  - a quantitative metric of a system's characteristics or performance of a particular attribute or subsystem.
  - an MOP typically measures a level of physical performance below that of the system as a whole.

Needs Validation Contd…[Operational Effectiveness]

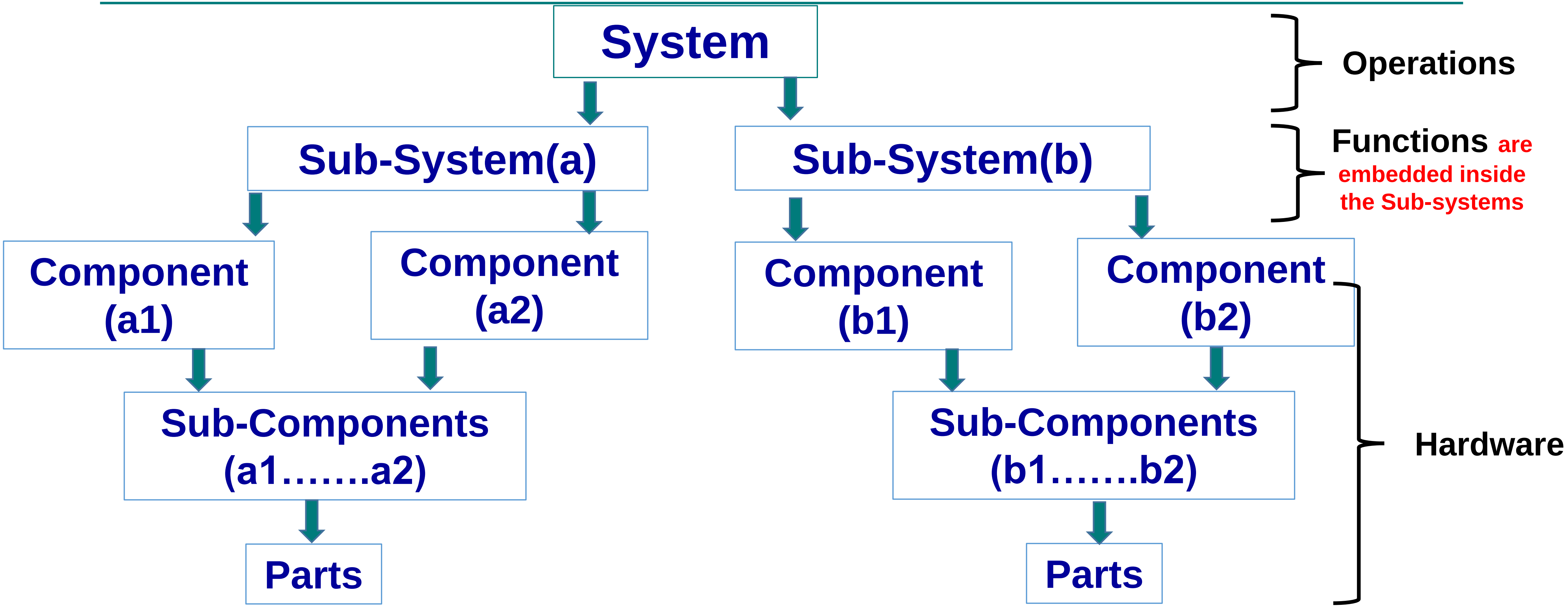
- **Technical Performance Measure (TPM)**
  - a quantitative metric of a system's characteristics or performance at the component level.
  - a TPM typically measures a level of physical performance below that of the sub-systemic level.

Needs Validation Contd…

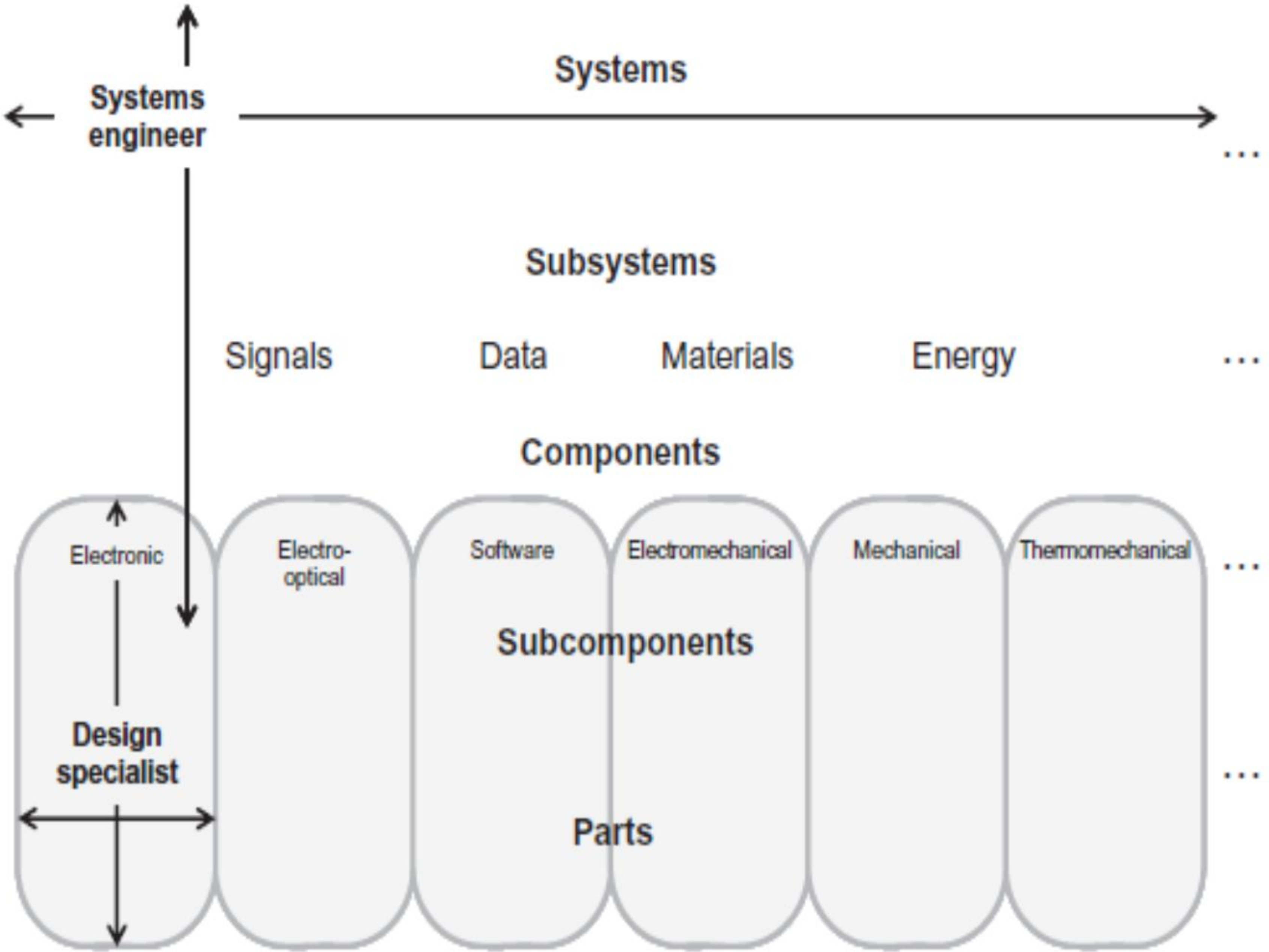


Levels of Effectiveness Measure

Structure of a System



System Design Hierarchy



Needs Validation Contd…

System Level Attributes

Engine Sound

Brightness of light

Car Speed

Stereo Sound

Needs Validation Contd…

Sub-system Level Attributes

Cylinder

Alternator

Horsepower

Player head unit

Needs Validation Contd…

Sub-system Level Attributes

Cylinder Horsepower

Alternator

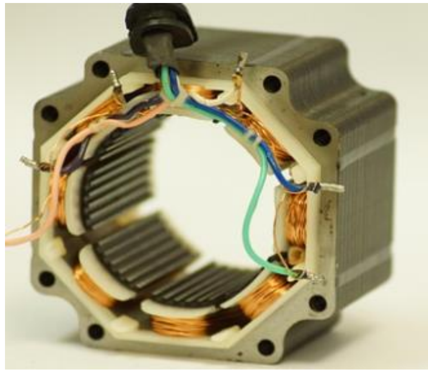
Player head unit

Needs Validation Contd…

Component Level Attributes

Piston

Rings



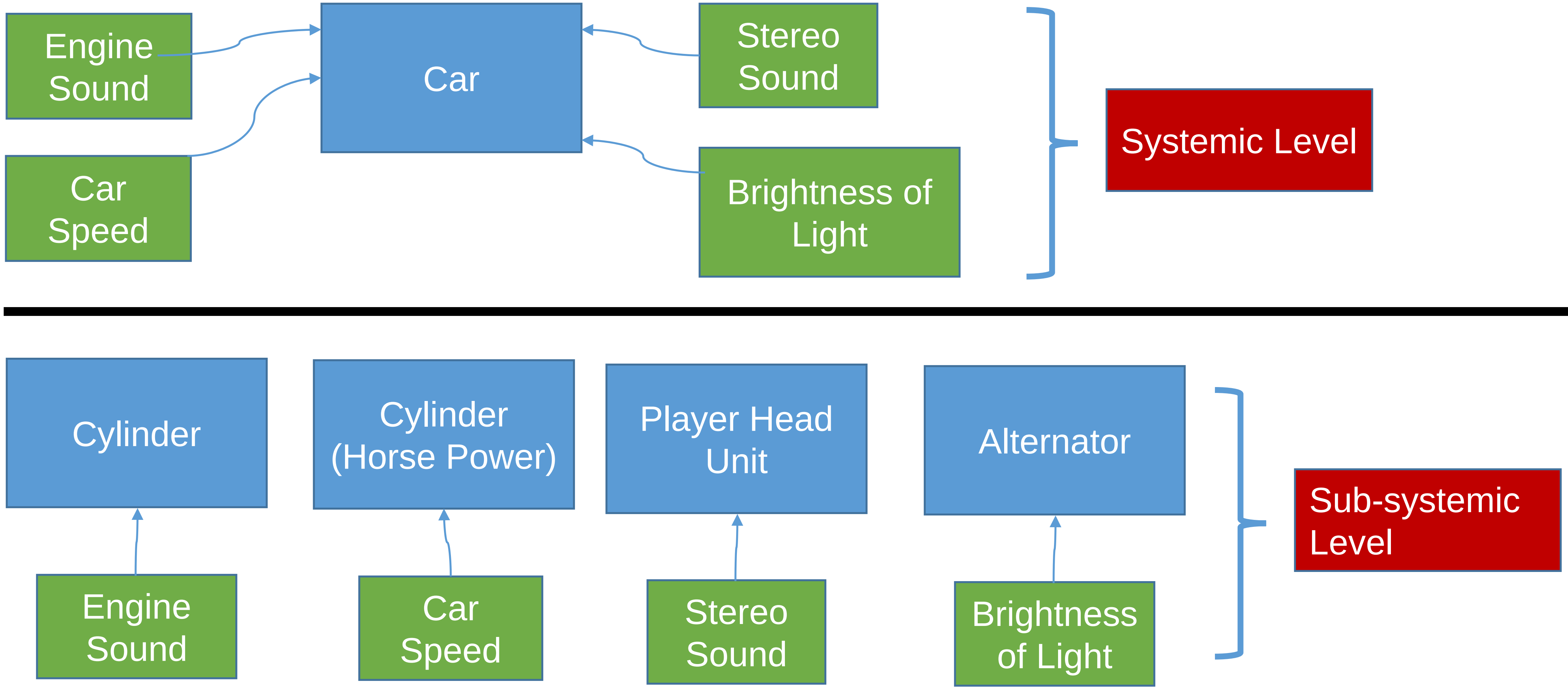
Stator

Rotor

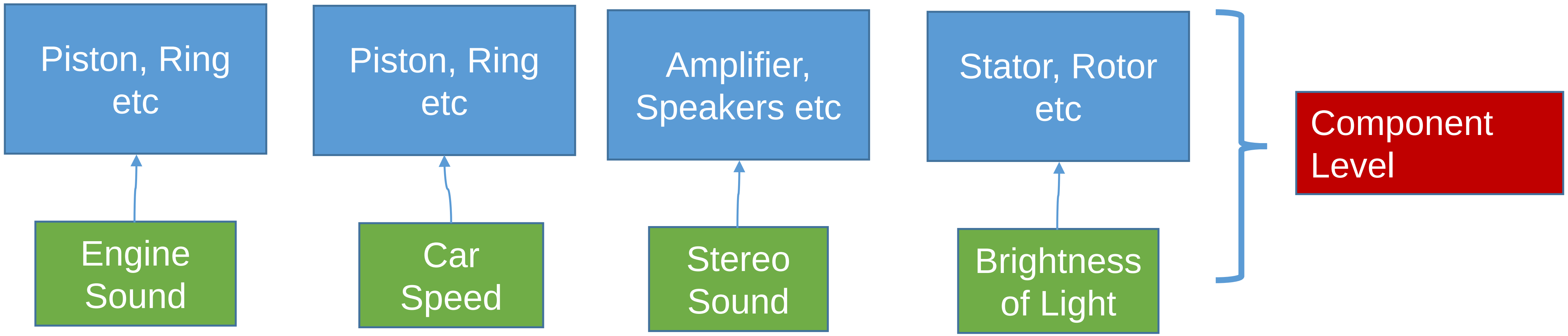
Amplifier

speakers

Needs Validation



A schematic view of NEEDS



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For further reading: Sections covered in the e-book include:

Systems Engineering Principles and Practice, Second Edition; Author: Alexander Kossiakoff et al.

- Part two
  - Chapter 6

System Engineering Management, Fifth Edition, Benjamin S. Blanchard & John E. Blyler,

- Chapter 2

Next lecture

Unit 2: Market Strategic Analysis and Competitors

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